SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Fundamentals of Data Science **COURSE CODE: DSA04**

COURSE OUTCOME COVERED

CO2: Apply data pre-processing and visualization techniques using Python libraries for effective data handling. (BL4)

QUESTIONS: 05 **Total Marks:100**

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I P.Vishwa Nandan(192473015) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Vishwa nandan

Annexure A

Analytical Problem Solving

1. Online Food Delivery Data Analysis
An online food delivery company wants to analyze order data to improve delivery performance and customer satisfaction. Explain how:
A. **Pandas** is used to read, clean, and filter order data
B. **Data preprocessing** is applied to handle missing delivery time, duplicate orders, and incorrect prices
C. **Matplotlib or Seaborn** is used to visualize delivery delays, top restaurants, and customer ratings

2. Bank Loan Approval Analysis
A bank wants to study customer loan applications and identify patterns related to loan approval and rejection. Explain how:
A. **Pandas DataFrame operations** are used to select income, loan amount, credit score, and approval status
B. **Preprocessing techniques** are used to handle missing income values, encode gender/employment type, and normalize loan amount
C. **Visualization** is used to compare approved and rejected applications based on income, credit score, and loan amount

3. Smart Energy Consumption Monitoring
A smart home system collects daily electricity consumption data from different appliances such as AC, fan, light, and refrigerator. Explain how:
A. **Pandas** is used to group electricity usage by appliance and date
B. **Data preprocessing** is used to remove abnormal readings, fill missing values, and calculate total power consumption
C. **Line charts and bar charts** are used to identify high-energy appliances and daily usage trends

4. Online Learning Platform Performance Analysis
An online learning platform wants to analyze student activity such as login time, video completion, quiz score, and assignment submission. Explain how:
A. **Pandas** is used to sort, filter, and rank students based on quiz scores and completion percentage
B. **Preprocessing** is applied to handle missing quiz scores, convert date formats, and create new columns such as performance category
C. **Seaborn and Matplotlib** are used to visualize student engagement, score distribution, and course completion status

5. Public Transport Passenger Data Analysis
A city transport department wants to analyze bus passenger data to improve route planning and reduce overcrowding. Explain how:
A. **Pandas** is used to analyze passenger count by route, stop, and time slot
B. **Data preprocessing** is used to clean invalid route numbers, handle missing passenger counts, and remove duplicate records
C. **Visualization techniques** are used to show peak travel hours, busiest routes, and passenger flow patterns

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

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1. Online Food Delivery Data Analysis

A. Using Pandas to Read, Clean, and Filter Order Data

- Pandas reads order data from CSV files using pd.read_csv().
- DataFrame operations help organize order details such as restaurant, delivery time, price, and customer rating.
- Filtering identifies delayed orders and orders from specific restaurants.
- Sorting helps identify restaurants with the highest ratings and longest delivery times.

B. Data Preprocessing

- **Missing values:** Fill missing delivery times using the median or remove records when appropriate.
- **Duplicate orders:** Identify and remove duplicate records using drop_duplicates().
- **Incorrect prices:** Check for negative prices and invalid values, then correct them using reliable source data.
- **Data validation:** Ensure delivery times and prices are within reasonable ranges.

C. Data Visualization Using Matplotlib or Seaborn

- **Bar chart:** Compare the average delivery time of different restaurants.
- **Histogram:** Display the distribution of delivery times.
- **Bar chart:** Identify restaurants with the highest number of orders.
- **Scatter plot:** Analyze the relationship between delivery time and customer ratings.

2. Bank Loan Approval Analysis

A. Using Pandas DataFrame Operations

- Pandas reads customer loan application data into a DataFrame.
- Select important columns such as income, loan amount, credit score, and approval status.
- Filter records to separate approved and rejected applications.
- Sort applications based on income, credit score, or loan amount.

B. Data Preprocessing

- **Missing income values:** Fill missing values using the median or another suitable method.

- **Categorical encoding:** Convert gender and employment type into numerical representations using encoding techniques.
- **Normalization:** Scale loan amounts to a common range using Min-Max normalization.
- **Data validation:** Check for invalid income, loan amounts, and credit scores.

C. Data Visualization

- **Bar chart:** Compare the number of approved and rejected applications.
- **Scatter plot:** Examine the relationship between income and loan amount.
- **Box plot:** Compare credit scores for approved and rejected applicants.
- **Histogram:** Analyze the income distribution of loan applicants.

3. Smart Energy Consumption Monitoring

A. Using Pandas to Group Electricity Usage

- Pandas reads electricity consumption data collected from smart home appliances.
- Group data by appliance name to calculate individual energy usage.
- Group data by date to calculate daily electricity consumption.
- Use groupby() and sum() to calculate total consumption for each appliance and date.

B. Data Preprocessing

- **Abnormal readings:** Identify unusually high or negative energy readings and investigate or remove invalid records.
- **Missing values:** Fill missing readings using suitable methods, such as time-based interpolation when appropriate.
- **Duplicate records:** Remove repeated sensor readings.
- **Total consumption:** Calculate the combined electricity usage of the AC, fan, lights, and refrigerator.

C. Data Visualization

- **Bar chart:** Compare electricity consumption across appliances.
- **Line chart:** Display daily electricity consumption trends.
- **Pie chart:** Show each appliance's percentage of total consumption.
- **Line chart:** Identify days with unusually high electricity usage.

4. Online Learning Platform Performance Analysis

A. Using Pandas to Sort, Filter, and Rank Students

- Pandas reads student activity data, including quiz scores, video completion, and assignment submissions.
- Filter students based on quiz scores and course completion percentages.
- Sort students according to their quiz scores.
- Rank students to identify high-performing students and those who need additional support.

B. Data Preprocessing

- **Missing quiz scores:** Fill missing values when appropriate or mark unattempted quizzes separately.
- **Date conversion:** Convert login and submission dates into standard datetime formats.
- **New columns:** Create a performance category based on quiz scores.
- **Data cleaning:** Remove duplicate records and validate completion percentages.

C. Data Visualization Using Seaborn and Matplotlib

- **Bar chart:** Compare student engagement across courses.
- **Histogram:** Display the distribution of quiz scores.
- **Heatmap:** Show relationships between student activity measures and performance.
- **Count plot:** Compare completed and incomplete courses.

5. Public Transport Passenger Data Analysis

A. Using Pandas to Analyze Passenger Counts

- Pandas reads passenger records containing route numbers, bus stops, timestamps, and passenger counts.
- Group passenger data by route to identify the busiest routes.
- Group data by stop to determine passenger demand at different locations.
- Group data by time slot to identify peak travel hours.

B. Data Preprocessing

- **Invalid route numbers:** Identify and correct invalid route numbers using official route records.
- **Missing passenger counts:** Fill missing values only when reliable information is available; otherwise, mark them as missing.
- **Duplicate records:** Remove duplicate passenger entries using appropriate identifiers.
- **Data validation:** Check for negative passenger counts and invalid timestamps.

C. Data Visualization

- **Bar chart:** Identify routes with the highest passenger counts.
- **Line chart:** Show passenger demand throughout the day.
- **Heatmap:** Compare passenger counts across routes and time slots.
- **Stacked bar chart:** Compare passenger numbers at different stops across selected routes.